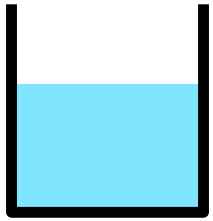
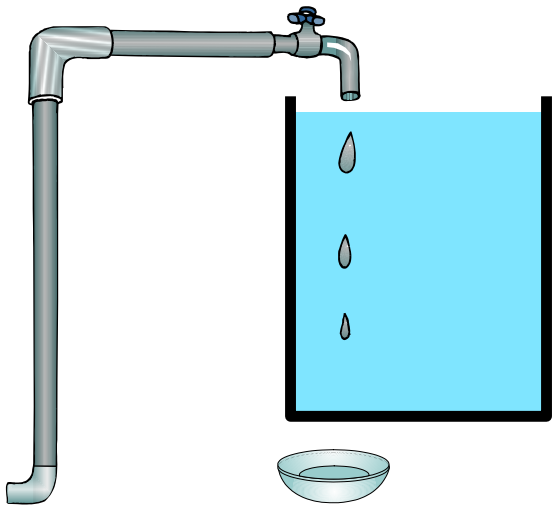
# Greatest Common Divisors



3 Gallon Jug



5 Gallon Jug

# Multiples and Divisors

*For nonzero b, b* is a *divisor* of a iff there is an integer k such that a=kb. Denoted as *b|a* or *b divides a* or *a is divided by b*

*For nonzero b, a* is a *multiple* of b iff there is an integer k such that a=kb

*b* is a *divisor* of a iff *a* is a *multiple* of b

Say a=5, b=10, then *a|b*.

Say a=0, b=30, then *b|a*

Say a=-3, b=11, then a does not divide b, denoted as $a \nmid b$,
and b does not divide a, denoted as $b \nmid a$

# Common Divisors

$c$ is a *common divisor* of $a$ and $b$ means $c|a$ and $c|b$.

the **greatest common divisor,** $gcd(a,b)$, is defined as the greatest common divisor of $a$ and $b$.

Say a=8, b=10, then 1,2 are common divisors, and gcd(8,10)=2.

Say a=10, b=30, then 1,2,5,10 are common divisors, and gcd(10,30)=10.

Say a=3, b=11, then the only common divisor is 1, and gcd(3,11)=1.

**Claim.** If $p$ is prime, and $p$ does not divide $a$, then $gcd(p,a) = 1$.

# The Quotient-Remainder Theorem

For $b > 0$ and any $a$, there are *unique* integers

$q$ ::= quotient($a$,$b$), $r$ ::= remainder($a$,$b$), such that

$$a = qb + r \quad \text{and} \quad 0 \le r < b.$$

We also say **q = a div b** and **r = a mod b**.

When b=2, there is a unique q such that

a=2q or a=2q+1.

$$q = \lfloor \frac{a}{2} \rfloor$$

When b=3, there is a unique q such that

a=3q or a=3q+1 or a=3q+2.

$$q = \lfloor \frac{a}{3} \rfloor$$

**Floor function** $\lfloor x \rfloor$ : the greatest integer that is $\le x$

# The Quotient-Remainder Theorem

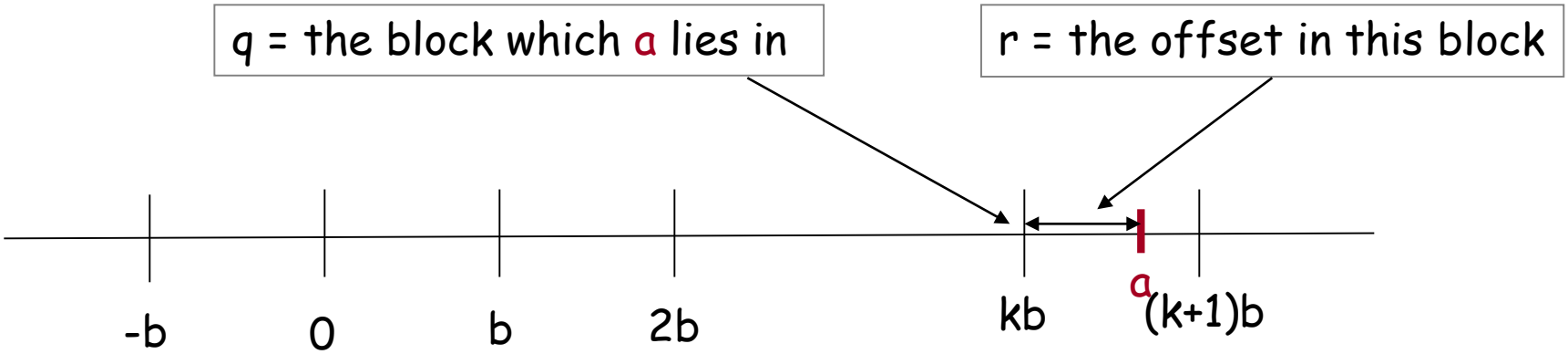
For $b > 0$ and any $a$, there are *unique* integers

$q$ ::= quotient($a$,$b$), $r$ ::= remainder($a$,$b$), such that

$$a = qb + r \quad \text{and} \quad 0 \le r < b.$$

Given any $b$, we can partition the integers into blocks of b numbers.

For any $a$, there is a unique "position" for this number.




Clearly, given $a$ and $b$, the numbers $q$ and $r$ are uniquely determined.

# Greatest Common Divisors

Given a and b, how to compute gcd(a,b)?

Maybe try every number? Not easy for large numbers…
Do we have a better way to do it?

Let's say $a \geq b > 0$.

gcd(a,b)

Let d=1

1. If d|a and d|b, then store d.
2. Let d=d+1
3. If $d \leq b$, return to 1.

   else the answer = maximum of all stored "d"s

# Greatest Common Divisors

Given a and b, how to compute gcd(a,b)?

Maybe try every number? Not easy for large numbers…
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Let's say $a \geq b > 0$.

1. If $a=kb$, then $\gcd(a,b)=b$, and we are done.
2. Otherwise, by the Division Theorem, $a = qb + r$ where $r>0$.

# Greatest Common Divisors

Let's say $a \geq b$.

1. If $a=kb$, then $\gcd(a,b)=b$, and we are done.
2. Otherwise, by the Division Theorem, $a = qb + r$ where $r>0$.

| | | |
|---|---|---|
| $a=12$, $b=8$ => $12 = 8 + 4$ | $\gcd(12,8) = 4$ | $\gcd(8,4) = 4$ |
| $a=21$, $b=9$ => $21 = 2\times9 + 3$ | $\gcd(21,9) = 3$ | $\gcd(9,3) = 3$ |
| $a=99$, $b=27$ => $99 = 3\times27 + 18$ | $\gcd(99,27) = 9$ | $\gcd(27,18) = 9$ |

Euclid: $\gcd(a,b) = \gcd(b,r)$!

# Euclid's GCD Algorithm

$a = qb + r$

Euclid: gcd(a,b) = gcd(b,r)!

Assumption: $a > b \geq 0$.

```
gcd(a,b)

if b = 0, then answer = a.

else

  write a = qb + r

  answer = gcd(b,r)
```

$$q = \lfloor \frac{a}{b} \rfloor \qquad r = a - qb$$

# Example 1

```
gcd(a,b)
if b = 0, then answer = a.
else
  write a = qb + r
  answer = gcd(b,r)
```

| | |
|---|---|
| GCD(102, 70) | 102 = 70 + 32 |
| = GCD(70, 32) | 70 = 2x32 + 6 |
| = GCD(32, 6) | 32 = 5x6 + 2 |
| = GCD(6, 2) | 6 = 3x2 + 0 |
| = GCD(2, 0) | |

Return value: 2.

# Example 2

```
gcd(a,b)
if b = 0, then answer = a.
else
  write a = qb + r
  answer = gcd(b,r)
```

| | |
|---|---|
| GCD(252, 189) | 252 = 1x189 + 63 |
| = GCD(189, 63) | 189 = 3x63 + 0 |
| = GCD(63, 0) | |

Return value: 63.

# Example 3

```
gcd(a,b)
if b = 0, then answer = a.
else
  write a = qb + r
  answer = gcd(b,r)
```

| | |
|---|---|
| GCD(662, 414) | 662 = 1x414 + 248 |
| = GCD(414, 248) | 414 = 1x248 + 166 |
| = GCD(248, 166) | 248 = 1x166 + 82 |
| = GCD(166, 82) | 166 = 2x82 + 2 |
| = GCD(82, 2) | 82 = 41x2 + 0 |
| = GCD(2, 0) | |

Return value: 2.

# Correctness of Euclid's GCD Algorithm

$$a = qb + r$$

Euclid: $\gcd(a,b) = \gcd(b,r)$

**When r = 0:**

Then $a = qb$, so $\gcd(a, b) = b$;

$r = 0$, so $\gcd(b, r) = \gcd(b, 0) = b$.

Therefore, $\gcd(a,b) = \gcd(b,r)$.

# Correctness of Euclid's GCD Algorithm

$a = qb + r$

Euclid: $\gcd(a,b) = \gcd(b,r)$

**When $r > 0$:**

Let d be a common divisor of b, r

$\Rightarrow b = k_1 d$ and $r = k_2 d$ for some $k_1$, $k_2$.

$\Rightarrow a = qb + r = qk_1 d + k_2 d = (qk_1 + k_2)d$ => d is a common divisor of a, b

Let d be a common divisor of a, b

$\Rightarrow a = k_3 d$ and $b = k_1 d$ for some $k_1$, $k_3$.

$\Rightarrow r = a - qb = k_3 d - qk_1 d = (k_3 - qk_1)d$ => d is a common divisor of b, r

So, {common factors of a, b} = {common factors of b, r}

$\Rightarrow \gcd(a, b) = \gcd(b, r)$.

# Is Euclid's GCD Algorithm fast?

Naive algorithm: try every number.

Without loss of generality: $a > b \geq 0$.

gcd(a,b)

Let d=1

1. If $d|a$ and $d|b$, then store d.
2. Let d=d+1
3. If $d \leq b$, return to 1.

   else the answer = maximum of all stored "d"s

So the running time is about b iterations.

# Is Euclid's GCD Algorithm fast?

Euclid's algorithm:

In two iterations, a, b are decreased by half. (why?)

$a = bq + r \geq b + r > 2r$

$\Rightarrow \gcd(a,b) = \gcd(b,r)$ where $r < a/2$

Similarly, if $b = rq' + r'$, then

$\gcd(b,r) = \gcd(r,r')$ where $r' < b/2$

Suppose the algorithm stops in 2k iterations.

Then $2^d \geq 2^{k-1}$. (Suppose $2^{d+1} \geq b > 2^d$)

So the running time is about $2\log_2 b$ iterations.

Exponentially faster!

# Linear Combination vs Common Divisor

Greatest common divisor

d is a common divisor of a and b if $d|a$ and $d|b$

gcd(a,b) = greatest common divisor of a and b

Smallest positive integer linear combination

d is an **integer linear combination** of a and b if $d=sa+tb$ for integers s,t.

spc(a,b) = **smallest positive** integer linear **combination** of a and b

**Theorem. gcd(a,b) = spc(a,b)**

# Linear Combination vs Common Divisor

**Theorem. gcd(a,b) = spc(a,b)**

For example, the greatest common divisor of 52 and 44 is 4.
And 4 is an integer linear combination of 52 and 44:

$$6 \cdot 52 + (-7) \cdot 44 = 4$$

Furthermore, no integer linear combination of 52 and 44 is equal to a smaller positive integer.

To prove the theorem, we will prove:

$\gcd(a,b) \le \mathrm{spc}(a,b)$

$\gcd(a,b) \mid \mathrm{spc}(a,b)$

$\gcd(a,b) \ge \mathrm{spc}(a,b)$

spc(a,b) divides a and b

# GCD ≤ SPC

**Claim.** If $d \mid a$ and $d \mid b$, then $d \mid sa + tb$ for any $s,t$.

*Proof.*

$d \mid a \Rightarrow a = dk_1$

$d \mid b \Rightarrow b = dk_2$

$sa + tb = sdk_1 + tdk_2 = d(sk_1 + tk_2)$

$\Rightarrow d \mid (sa+tb)$

---

GCD | SPC

Let $d = \gcd(a,b)$. By definition, $d \mid a$ and $d \mid b$.

Let $f = \mathrm{spc}(a,b) = sa+tb$

According to the claim, $d \mid f$. So $\gcd(a,b) \leq \mathrm{spc}(a,b)$.

# GCD ≥ SPC

We will prove that spc(a,b) is actually a common divisor of a and b.

First, show that spc(a,b) | a.

1. By the Division Theorem (since a ≥ spc(a,b)),

   $$a = q \times spc(a,b) + r \qquad \text{and} \qquad spc(a,b) > r \geq 0$$

2. Let spc(a,b) = sa + tb.
3. Then r = a – q x spc(a,b) = a – q x (sa + tb) = (1-qs)a + qtb.
4. So r is an integer linear combination of a and b with spc(a,b) > r.
5. This is only possible when r = 0.

Similarly, spc(a,b) | b.

Thus, spc(a,b) divides both a and b, which follows spc(a,b) ≤ gcd(a,b).

# Application of the Theorem

**Theorem.** $\gcd(a,b) = \mathrm{spc}(a,b)$

**Lemma.** If $\gcd(a,b)=1$ and $\gcd(a,c)=1$, then $\gcd(a,bc)=1$.

By the **Theorem**, there exist s,t,u,v such that

$$sa + tb = 1$$
$$ua + vc = 1$$

So $(sa + tb)(ua + vc) = 1$

Expanding LHS gives

$$saua + savc + tbua + tbvc = 1$$

$\Rightarrow (sau + svc + tbu)a + (tv)bc = 1$

This implies $\mathrm{spc}(a,bc)=1$. By **Theorem**, we have $\gcd(a,bc)=1$.

# Prime Divisibility

**Theorem.** $\gcd(a,b) = \mathrm{spc}(a,b)$

***Lemma.*** $p$ prime and $p|ab$ implies $p|a$ or $p|b$.

*proof.* W.l.o.g, assume p does not divide a. Then $\gcd(p,a)=1$.

So by **Theorem**, there exist s and t such that

$$sa + tp = 1$$

$$\underbrace{(sa)b}_{p|ab} + \underbrace{(tp)b}_{p|p} = b$$

Hence $p|b$

***Corollary.*** If $p$ is prime, and $p| a_1a_2\cdots a_m$ then $p|a_i$ for some $i$.

# Fundamental Theorem of Arithmetic

Every integer $n>1$ has a *unique* factorization into primes:

$$p_0 \le p_1 \le \cdots \le p_k,$$

$$n = p_0\, p_1 \cdots p_k.$$

$p_i$ might duplicate

*Example:*

$61394323221 = 3\cdot3\cdot3\cdot7\cdot11\cdot11\cdot37\cdot37\cdot37\cdot53$

# Unique Factorization

*Theorem.* There is a unique factorization.

*Proof.* Suppose there is a number with two different factorizations.

By Well Ordering Principle, we choose the smallest such $n > 1$:

$$n = p_1 \cdot p_2 \cdots p_k = q_1 \cdot q_2 \cdots q_m$$

Since n is smallest, we must have that $p_i \neq q_j$ all $i,j$

(Otherwise, we can obtain a smaller counterexample.)

Since $p_1 | n = q_1 \cdot q_2 \cdots q_m$, so by Corollary $p_1 | q_i$ for some i.

Since both $p_1$ , $q_i$ are prime numbers, we must have $p_1 = q_i$.

contradiction!

# Extended GCD Algorithm

How can we write gcd(a,b) as an integer linear combination?

This can be done by extending the Euclidean algorithm.

Example: a = 259, b=70

| | |
|---|---|
| $259 = 3 \cdot 70 + 49$ | $49 = a - 3b$ |
| $70 = 1 \cdot 49 + 21$ | $21 = 70 - 49$ |
| | $21 = b - (a-3b) = -a+4b$ |
| $49 = 2 \cdot 21 + 7$ | $7 = 49 - 2 \cdot 21$ |
| | $7 = (a-3b) - 2(-a+4b) = \underline{3a - 11b}$ |
| $21 = 7 \cdot 3 + 0$ | done, gcd = 7 |

# Extended GCD Algorithm

Example: $a = 899$, $b=493$

$899 = 1 \cdot 493 + 406$ so $406 = a - b$

$493 = 1 \cdot 406 + 87$ so $87 = 493 - 406$

$= b - (a-b) = -a + 2b$

$406 = 4 \cdot 87 + 58$ so $58 = 406 - 4 \cdot 87$

$= (a-b) - 4(-a+2b) = 5a - 9b$

$87 = 1 \cdot 58 + 29$ so $29 = 87 - 1 \cdot 58$

$= (-a+2b) - (5a-9b) = \underline{-6a + 11b}$

$58 = 2 \cdot 29 + 0$ done, gcd = 29

# Die Hard

Simon says: On the fountain, there are 2 jugs, one is 5-gallon and the other is 3-gallon. Fill one with exactly 4 gallons of water and place it on the scale then the timer will stop. You must be precise; one ounce more or less will result in detonation. If you're still alive in 5 minutes, we'll speak.

# Die Hard

**Bruce:** Wait, wait a second. I don't get it. Do you get it?

**Samuel:** No.

**Bruce:** Get the jugs. Obviously, we can't fill the 3-gallon jug with 4 gallons of water.

**Samuel:** Obviously.

**Bruce:** All right. I know, here we go. We fill the 3-gallon jug exactly to the top, right?

**Samuel:** Uh-huh.

**Bruce:** Okay, now we pour this 3 gallons into the 5-gallon jug, giving us exactly 3 gallons in the 5-gallon jug, right?

**Samuel:** Right, then what?

**Bruce:** All right. We take the 3-gallon jug and fill it a third of the way...

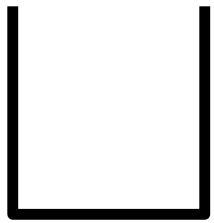
**Samuel:** No! He said, "Be precise." Exactly 4 gallons.

**Bruce:** Sh - -. Every cop within 50 miles is running his a** off and I'm out here playing kids games in the park.

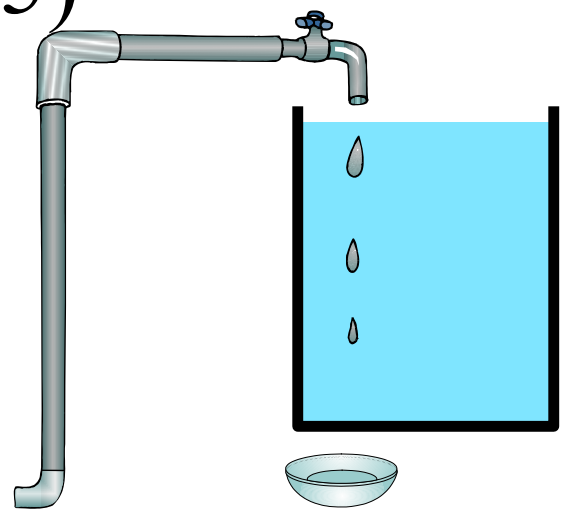
**Samuel:** Hey, you want to focus on the problem at hand?

# Die Hard

Start with empty jugs: (0,0)
Fill the big jug: (0,5)



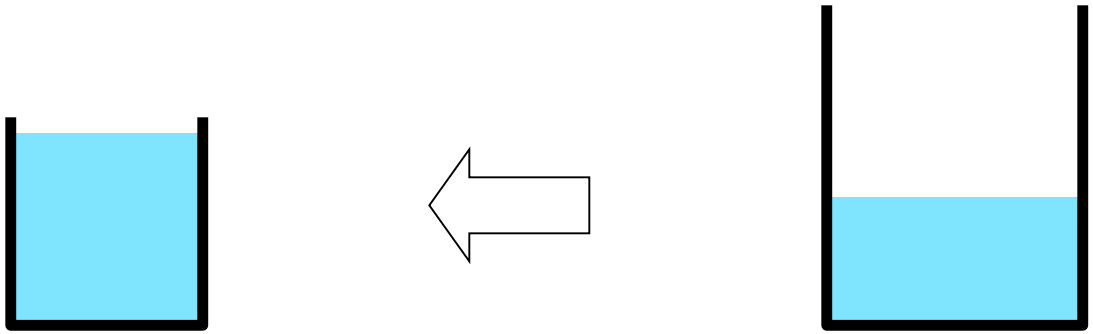
3-Gallon Jug



5-Gallon Jug

# Die Hard

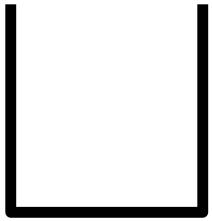
Pour from big to little: (3,2)



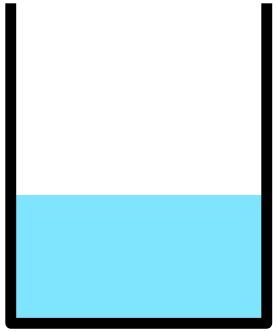
3-Gallon Jug 5-Gallon Jug

# Die Hard

Empty the little: (0,2)



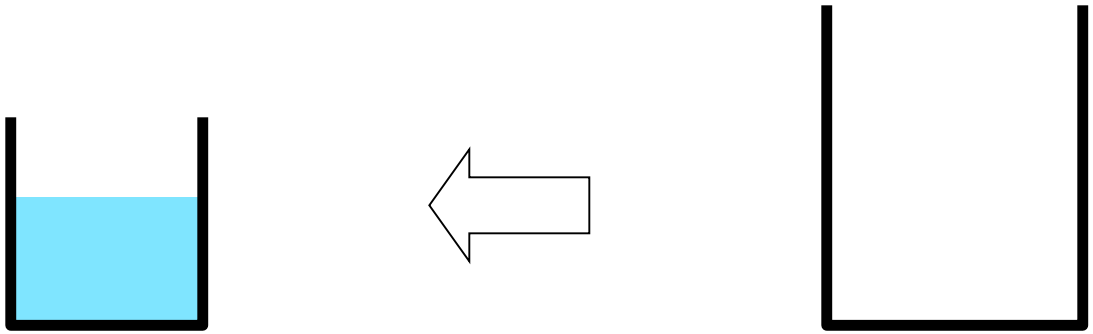
3-Gallon Jug



5-Gallon Jug

# Die Hard

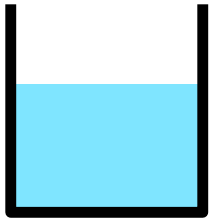
Pour from big to little: (2,0)



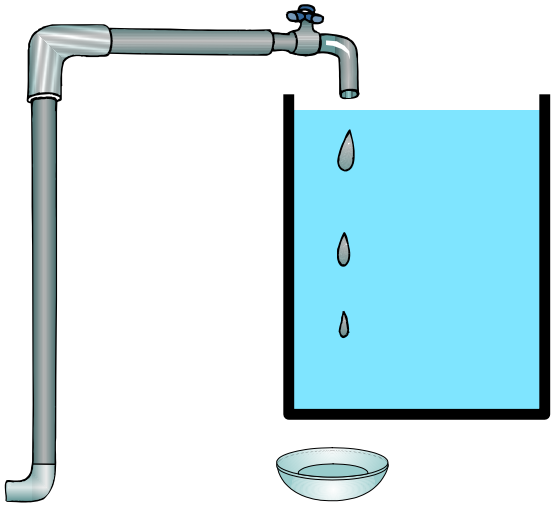
3-Gallon Jug 5-Gallon Jug

# Die Hard

Fill the big jug: (2,5)



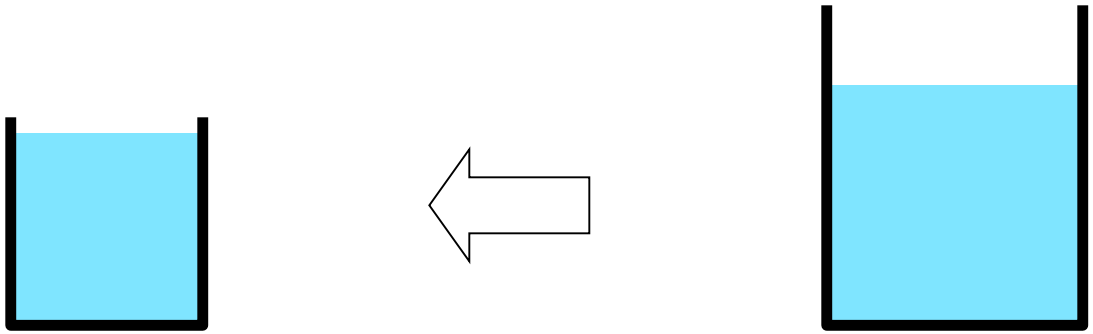
3-Gallon Jug



5-Gallon Jug

# Die Hard

Pour from big to little: (3,4)

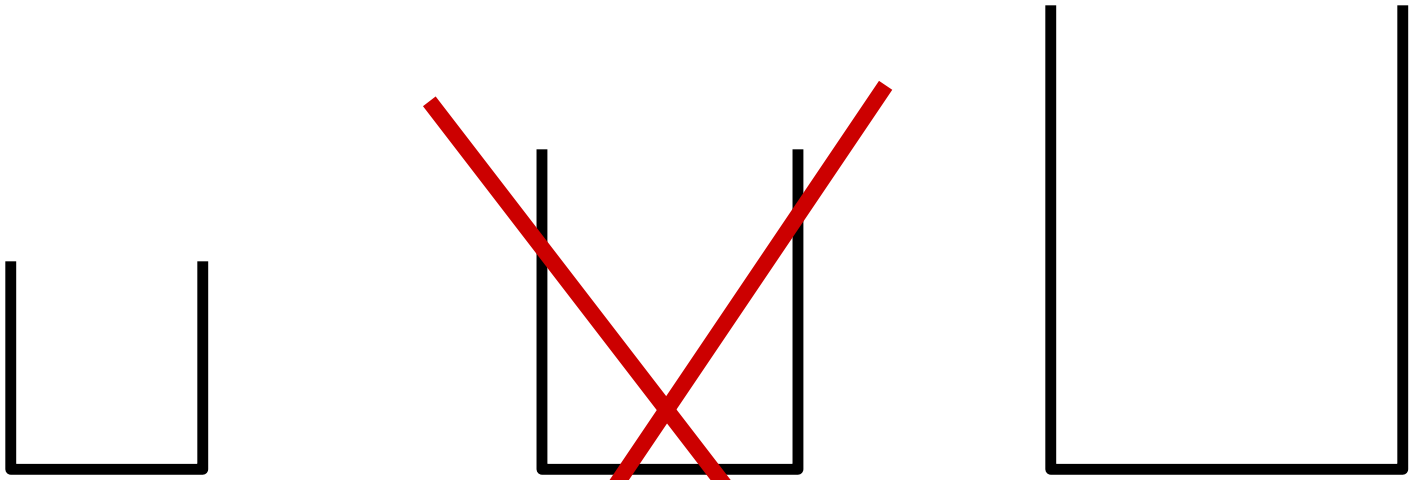


3-Gallon Jug 5-Gallon Jug

Done!!

# Die Hard

What if you have a 9 gallon jug instead?

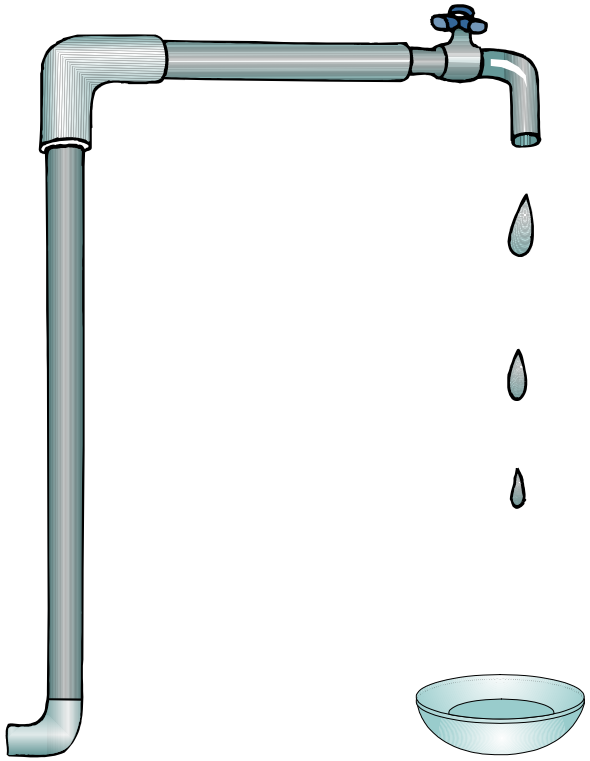


3 Gallon Jug ~~5 Gallon Jug~~ 9 Gallon Jug

Can you do it? Can you prove it?

# Die Hard

Supplies:



Water

3-Gallon Jug

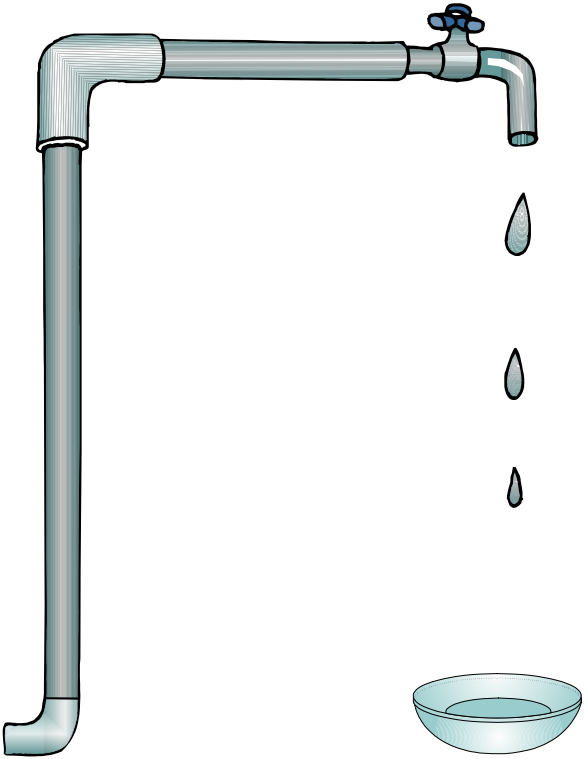
9-Gallon Jug

# Invariant Method

**Invariant:** the number of gallons in each jug is a multiple of 3.

i.e., $3|L$ and $3|B$ (3 divides both $L$ and $B$)

**Corollary.** It is impossible to have exactly 4 gallons in one jug.



Bruce Dies!

# Generalized Die Hard

Can Bruce form 3 gallons using 21 and 26-gallon jugs?

This question is not so easy to answer without number theory.

# General Solution for Die Hard

**Invariant in Die Hard Transition:**

Suppose that we have water jugs with capacities B and L. Then the amount of water in each jug is always an integer linear combination of B and L.

**Lemma.** gcd(a, b) divides any integer linear combination of a and b.

Let d = gcd(a,b). Then

$$d|a \quad \text{and} \quad d|b$$

So $d|ax+by$.

**Corollary.** The amount of water in each jug is a multiple of gcd(a,b).

# General Solution for Die Hard

**Corollary.** The amount of water in each jug is a multiple of gcd(a,b).

Given jug of 3 and jug of 9, is it possible to have exactly 4 gallons in one jug?

NO, because gcd(3,9)=3, and 4 is not a multiple of 3.

Given jug of 21 and jug of 26, is it possible to have exactly 3 gallons in one jug?

gcd(21,26)=1, and 3 is a multiple of 1,
so this means possible??

**Theorem.** Given water jugs of capacity a and b with $a \leq b$,
it is possible to have exactly k ($\leq$ b) gallons in one jug
if and only if k is a multiple of gcd(a,b).

# General Solution for Die Hard

**Theorem.** Given water jugs of capacity a and b with $a \le b$, it is possible to have exactly k ($\le$ b) gallons in one jug if and only if k is a multiple of gcd(a,b).

Given jug of 21 and jug of 26, is it possible to have exactly 3 gallons in one jug?

$$\gcd(21,26) = 1$$

$$\Rightarrow 5 \times 21 - 4 \times 26 = 1$$

$$\Rightarrow 15 \times 21 - 12 \times 26 = 3$$

Repeat 15 times:

1. Fill the 21-gallon jug.
2. Pour all the water in the 21-gallon jug into the 26-gallon jug. Whenever the 26-gallon jug becomes full, empty it out.

# General Solution for Die Hard

$$15 \times 21 - 12 \times 26 = 3$$

Repeat 15 times:

1. Fill the 21-gallon jug.
2. Pour all the water in the 21-gallon jug into the 26-gallon jug.

   Whenever the 26-gallon jug becomes full, empty it out.

**Claim.** There must be exactly 3 gallons left after this process.

1. Totally we have filled 15x21 gallons.
2. We pour out t multiple of 26 gallons.
3. The 26 gallon jug can only hold the volume between 0 and 26.
4. So t must be 12.
5. And there is exactly 3 gallons left.

# General Solution for Die Hard

Given two jugs with capacity $A$ and $B$ with $A \leq B$, the target is $C$.

If gcd(A,B) does not divide C, then it is impossible.

Otherwise, compute $C = sA + tB$. (We can always make $s > 0$.)

Repeat $s$ times:

1. Fill the A-gallon jug.
2. Pour all the water in the A-gallon jug into the B-gallon jug.

   Whenever the B-gallon jug becomes full, empty it out.

The B-gallon jug will be emptied exactly $t$ times.

After that, there will be exactly C gallons in the B-gallon jug.